



Coordinate Compression

- Harsh Gupta

Goal



To understand:

- Understand Coordinate Compression
- Using Difference Array with Coordinate Compression



Problem 1

Given an array **arr**, containing **integers** from **1** to **10^{12}** , **compress** them to numbers ranging from **1** to **10^6**

n numbers $\rightarrow n \leq 2e5 \rightarrow 2 \times 10^5$

$a_i \leq 1e12 \rightarrow 10^{12}$

$a_1 < a_2$

$\rightarrow 0 \rightarrow 1e6$

$a_i < a_j$

$0 \rightarrow 10^{12}$

$a_i < a_j$

$i < j$

$a_i < a_j$

$1 \quad 10^{12}$

$\textcircled{1} \quad \textcircled{2}$
 $\downarrow \quad \downarrow$
 $0 \quad 0$
 $\textcircled{1e6-1} \quad \textcircled{1e6+1}$

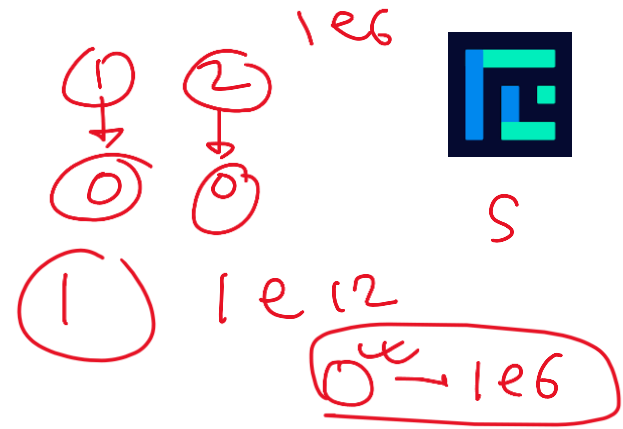


subtraction X

Division X

modular X

Log X



(2) (40)

[ⁱ(2), 1e11, 1e12, 1000, ^j(40)]



After
sorting

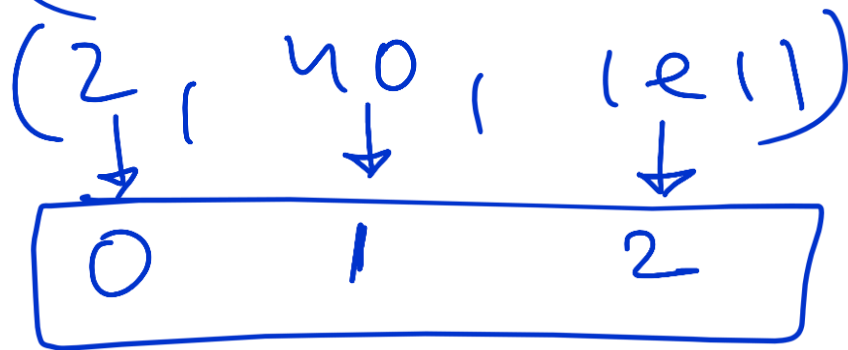
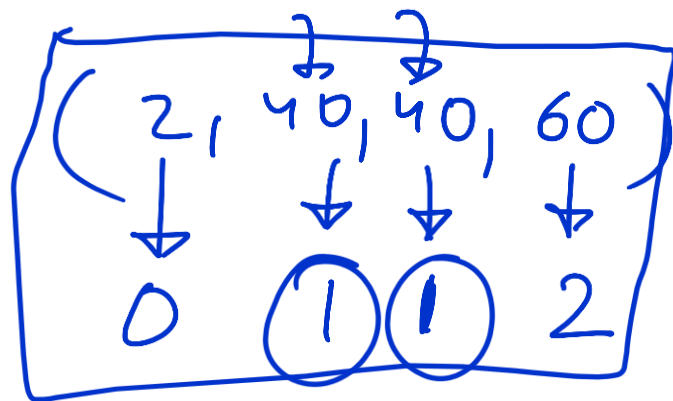
[2, 40, 1000, 1e11, 1e12]

[0, 1, 2, 3, 4]

2e5

[0, 1e6]

2e5-1
< 2e5
2e5-1



Problem 2



Snuke Prime - https://atcoder.jp/contests/abc188/tasks/abc188_d



N ~~programs~~ services

i^{th} service

(C)

a_i	b_i	c_i
-------	-------	-------

2

4

(6)

day 2

day 3

day 4

$\rightarrow 1)$ $\downarrow 2$ $\downarrow 4$ $\downarrow 6$
 $\checkmark 2)$ $\checkmark 4$ $\checkmark 5$ $\checkmark 5$

$C = 10$



day 2
6

day 3
~~6~~
10

day 4
~~6~~
~~5~~
10

days day 6
5

- | | | | | |
|---------------|-----------------|-----------------|---|---|
| | a_i | b_i | c_i | |
| ✓ service 1 → | 2 | 4 | <u>6</u> | $\times \times (10)$
<div style="border: 1px solid blue; padding: 5px; display: inline-block;"> $C = 10$ </div> |
| ✓ service 2 → | 4 | 6 | <u>5</u> | |
| | day 2 | day 3 | day 4 | day 5 |
| | 6
↓
✓ (6) | 6
↓
✓ (6) | 6 ✓
5 ✓
↓
(11) → 10 | 5
↓
(5) |
| | | | | 5
↓
(5) ✓ |

$$\times \underline{a_i, b_i \leq 1e6} \times$$

days
↓
index



(c_i)

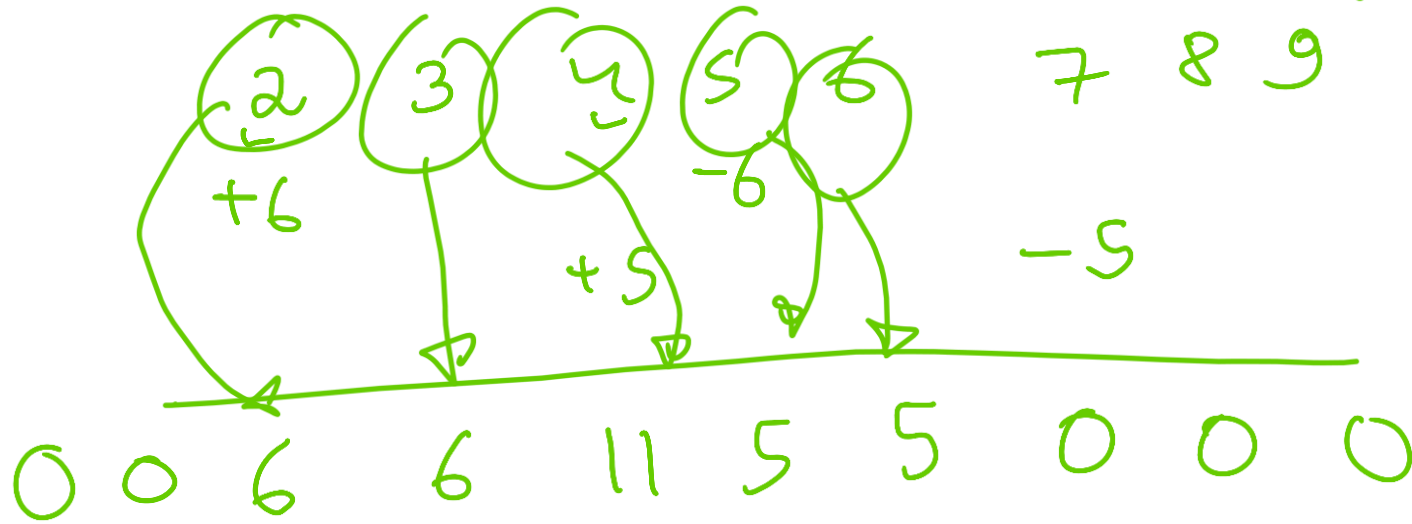
$$\begin{array}{ccccccc} \underbrace{a_i} & \times & \times \times & \underbrace{b_i} & b_{i+1} \\ + \underbrace{c_i} & & & & - \underbrace{c_i} \end{array}$$

day 2, 4, 5 ✓ (10)

4, 6, 5

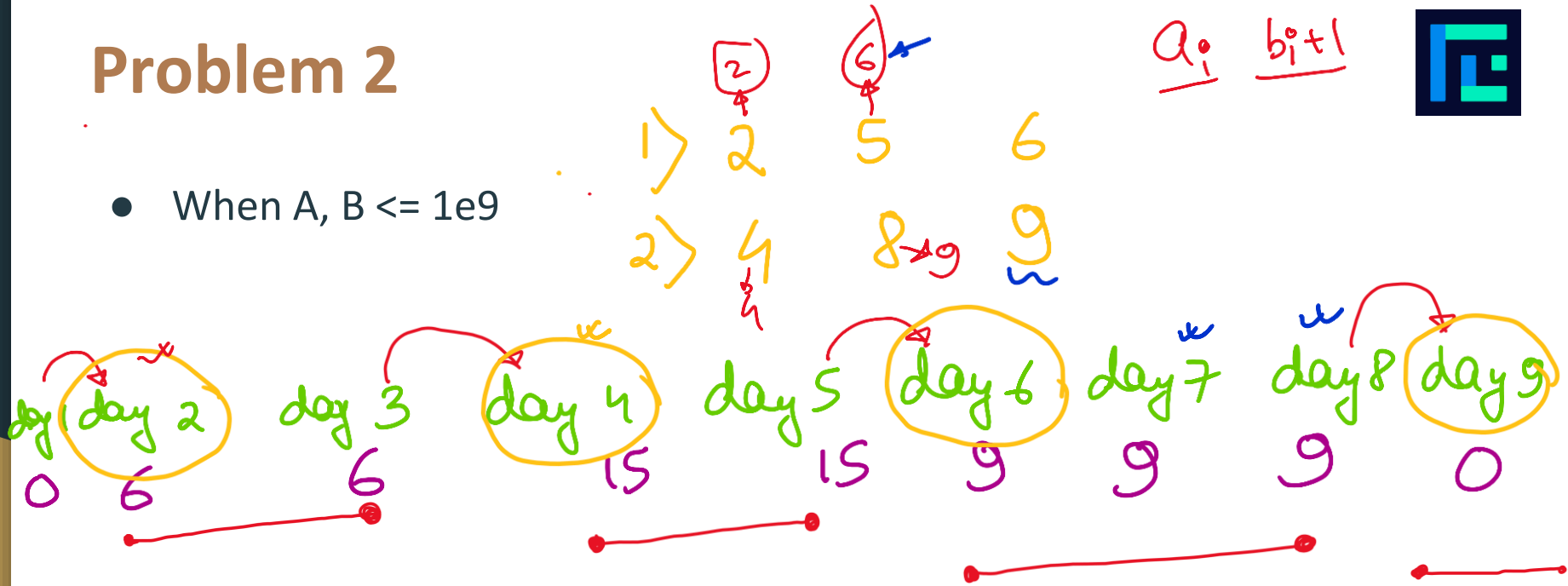
$10^6 + 1$. day 1 → day 106

every index represents a day.



Problem 2

- When $A, B \leq 1e9$



Info 1 → day 7 & day 8 cost will be equal to day 6.
Info 2 → Cost on day 6.

Q: $b_i + 1 \rightarrow$ important points

days

day 8



2 5 6
 4 8 9
 8 11 9



Prefix sum



day 2 day 3 day 4 days day 6 day 9

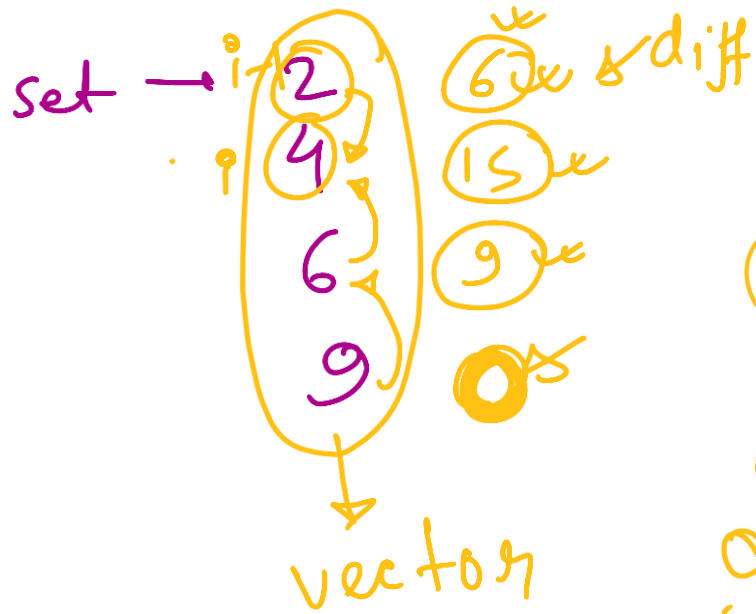


C (circled)

$$C = 10$$

day 8 6

$$6 + 6 + 10 + 10 + 9 + 9 + 9$$



2, 4, 6, 9



$$4 - 2 = 6$$

$$6 - 4 = 15$$

$$9 - 6 = 9$$

Steps for using Coordinate Compression



1. Figure out the **important points**.
2. **Map** out the important points.
3. Solve question with only those important points.
4. Figure out the answer for all **remaining points**.